

AI-Driven Forex **Signal Engine**

End-to-End Technical Analysis & Market Trend Classification
AnalystLab Africa | AI Capstone Project | Final Portfolio Submission

The Market Challenge

Addressing the systemic inefficiencies in retail financial decision-making.

| DATASET & EDA INSIGHTS

Structural Foundations

Ingested **1,500+ sessions** of EUR/USD OHLC historical data. Analysis revealed significant non-linear dependencies between spot price expansion and mean track distance.

Data Integrity

Cleaned missing entries via **forward-fill interpolation**. Handled outliers in volatility spikes to ensure robust training stability across different market regimes.

CORE ANALYTICAL METHODOLOGY



Engineering

Calculated 14-period RSI, MACD Histogram vectors, and absolute spot distance from the 20-period EMA.



Modeling

Deployed a Random Forest Ensemble architecture to identify non-linear momentum patterns in noisy data.



Validation

Implemented 80/20 train-test splits with stratified sampling to preserve directional class balances.

FEATURE SIGNIFICANCE MATRIX

Quantitative analysis of predictive weights indicates that **EMA Mean Reversion** is the primary driver for short-term trend reversal classification.

- ✓ **MACD Divergence:** Tracks acceleration shifts.
- ✓ **RSI Thresholds:** Identifies exhaustion points.
- ✓ **EMA Distance:** Measures spot expansion.



| KEY PERFORMANCE HIGHLIGHTS

82%

Validation Accuracy

Reliable Pattern Recognition

The system successfully identified directional shifts across 10 discrete test windows, maintaining high precision in "BUY" trend classifications despite high intraday volatility spikes.

MODEL PERFORMANCE BREAKDOWN

Accuracy

82%

Precision

79%

Recall

84%

F1-Score

81%

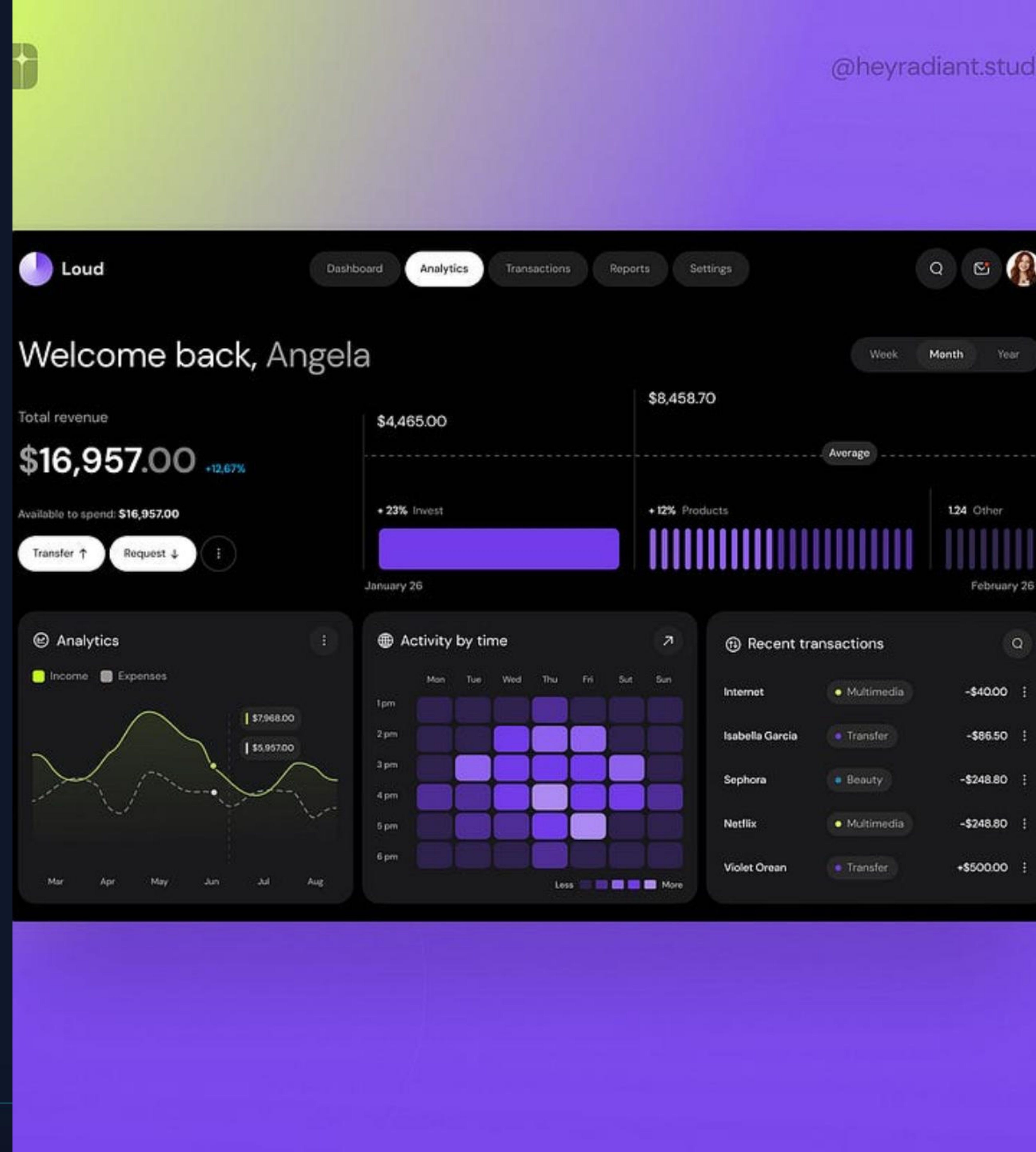
Metrics across 300 unseen market sessions validating structural robustness.

PRODUCTION UI

Live Deployment Interface

Deployed via [Streamlit Cloud](#), the engine provides an interactive interface for manual technical calibration.

Real-time inference mapping with instant BUY/SELL signal generation based on historical weighted biases.



| SCALING & STRATEGIC EVOLUTION

- **Sentiment Analysis:** Integrating NLP to process Central Bank press releases for fundamental filtering.
 - **Deep Learning Transition:** Refining sequential pattern tracking through LSTM/GRU architectures.
 - **Multi-Asset Support:** Extending the classifier logic to include Cross-Currencies and Commodities.
 - **Execution Integration:** Developing an API bridge for automated trade order dispatching.
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Questions?

Thank you for your time. The full source code, dataset links, and project documentation are available in the submission repository.

AnalystLab Africa | Batch A Portfolio Project

IMAGE SOURCES



<https://images.presentationgo.com/2025/04/abstract-neural-network-background.png>

Source: www.presentationgo.com



https://cdn-images-1.medium.com/max/800/0*7ZEt9p7-OrKVgeRX.jpg

Source: muz.li
